

Question-2

Given expression level of a gene follows a normal distribution

$$f(x) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{8}}$$

Given four expression values

$$x_1 = 10, x_2 = 13, x_3 = 15, x_4 = 20$$

Calculating MLE - Maximum Likelihood Estimation to estimate θ

The variance $\sigma^2 = 4$ $\sigma = 2$

the constant $\frac{1}{2\sqrt{2\pi}}$ is the normalization factor,

where $\sigma = 2$ and $\frac{1}{\sqrt{2\pi\sigma^2}} = \frac{1}{\sqrt{2\pi \cdot 4}} = \frac{1}{\sqrt{8\pi}} = \frac{1}{2\sqrt{2\pi}}$

* Likelihood function. ($L(\theta)$)

$$L(\theta) = f(x_1) \cdot f(x_2) \cdot f(x_3) \cdot f(x_4)$$

So, substituting the values

$$L(\theta) = \left(\frac{1}{2\sqrt{2\pi}} e^{-\frac{(10-\theta)^2}{8}} \right) \cdot \left(\frac{1}{2\sqrt{2\pi}} e^{-\frac{(13-\theta)^2}{8}} \right) \cdot \left(\frac{1}{2\sqrt{2\pi}} e^{-\frac{(15-\theta)^2}{8}} \right) \cdot \left(\frac{1}{2\sqrt{2\pi}} e^{-\frac{(20-\theta)^2}{8}} \right)$$

$$L(\theta) = \left(\frac{1}{2\sqrt{2\pi}}\right)^4 \cdot e^{-\frac{(10-\theta)^2}{8}} \cdot e^{-\frac{(13-\theta)^2}{8}} \cdot e^{-\frac{(15-\theta)^2}{8}} \cdot e^{-\frac{(20-\theta)^2}{8}}$$

using the property of exponents ($e^a \cdot e^b = e^{a+b}$)

$$L(\theta) = \left(\frac{1}{2\sqrt{2\pi}}\right)^4 \cdot e^{-\frac{(10-\theta)^2 + (13-\theta)^2 + (15-\theta)^2 + (20-\theta)^2}{8}}$$

* Simplify using log-likelihood

Maximizing $L(\theta)$ is equivalent to maximizing its natural logarithm

$$l(\theta) = \ln(L(\theta)) = \ln\left[\left(\frac{1}{2\sqrt{2\pi}}\right)^4 \cdot e^{-\frac{\sum_{i=1}^4 (x_i - \theta)^2}{8}}\right]$$

$$l(\theta) = \ln\left(\left(\frac{1}{2\sqrt{2\pi}}\right)^4\right) + \ln\left(e^{-\frac{\sum_{i=1}^4 (x_i - \theta)^2}{8}}\right)$$

$$l(\theta) = 4 \ln\left(\frac{1}{2\sqrt{2\pi}}\right) - \frac{\sum_{i=1}^4 (x_i - \theta)^2}{8}$$

Since $\frac{1}{2\sqrt{2\pi}}$ is a constant, the first term does not depend on θ . To maximize $l(\theta)$, we need to minimize the second term

$$-\frac{\sum_{i=1}^4 (x_i - \theta)^2}{8} \quad \text{or} \quad \sum_{i=1}^4 (x_i - \theta)^2$$

now defining the sum of squared differences:

$$S(\theta) = (10 - \theta)^2 + (13 - \theta)^2 + (15 - \theta)^2 + (20 - \theta)^2$$

we need to find θ that minimizes $S(\theta)$

$$\frac{dS}{d\theta} = -2(10 - \theta) - 2(13 - \theta) - 2(15 - \theta) - 2(20 - \theta) = 0$$

$$-2(10 - \theta + 13 - \theta + 15 - \theta + 20 - \theta) = 0$$

$$-2(10 + 13 + 15 + 20 - 4\theta) = 0$$

$$10 + 13 + 15 + 20 - 4\theta = 0$$

$$58 - 4\theta = 0$$

$$4\theta = 58$$

$$\theta = \frac{58}{4} = 14.5$$

verifying the maximum derivative of $S(\theta)$

$$\frac{d^2S}{d\theta^2} = 2 + 2 + 2 + 2 = 8$$

since $\frac{d^2S}{d\theta^2} = 8 > 0$, the critical point at $\theta = 14.5$ is a minimum of $S(\theta)$, means in MLE

$$\text{So, MLE for } (\theta) \text{ is } \theta = \frac{10 + 13 + 15 + 20}{4} = \frac{58}{4} = \underline{\underline{14.5}}$$